

Chapter 56: Attention as a Scarce Resource: Exploring the Determinants of Attention Consumption among Gen-Z

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Abstract

In today's digital economy, attention has emerged as a scarce resource, with online platforms competing continuously to capture and retain users' cognitive engagement. Against this backdrop, employing a multivariate regression model, our study examines how various determinants, such as attention fragmentation, fear of missing out, and attention awareness, influence the attention consumption of Gen-Z. To achieve the objective, primary data is collected from 150 respondents in Assam. Findings show that attention fragmentation has a positive and significant effect on attention consumption, whereas fear of missing out and attention awareness do not exhibit such significance. By viewing attention through the lens of microeconomic scarcity, these findings suggest that addressing attention fragmentation should be prioritised over awareness-based interventions and support emerging global efforts, such as the European Union's Digital Services Act, that advocate greater transparency in AI-driven recommender systems and more human-centred digital platform design.

Keywords: Attention Consumption, Attention Fragmentation, Fear of Missing Out (FOMO), Digital Media, Gen-Z.
JEL Classification: D83, D91, L86, O33.

1. Introduction

'A wealth of information creates a poverty of attention.' - Herbert A. Simon (1971)

In today's world, the way people communicate, access information, and spend their leisure time has been completely transformed by the widespread use of digital media. For Gen-Z, media platforms like Instagram, YouTube, and WhatsApp have become essential parts of daily lives. As the content on these platforms continues to grow rapidly, individuals face a constant influx of information, social connections, and entertainment options that compete for their limited attention. In this regard, as noted by Herbert Simon (1971), a wealth of information inevitably creates a poverty of attention. This is because in an information-rich digital environment, individuals have a fundamentally limited capacity to process and adapt. Consequently, attention has become a scarce and highly valuable cognitive resource in our contemporary economy.

From a microeconomic perspective, scarcity implies that individuals must allocate limited resources among competing alternatives (Loewenstein & Wojtowicz, 2025). While economics traditionally considers resources such as income, labour, and time as scarce resources; attention also satisfies the characteristics of a scarce resource because it cannot be expanded indefinitely, and allocating it to one activity inevitably reduces its availability for others. Consequently, attention carries an opportunity cost, making every decision to engage with digital media simultaneously a decision to forego alternative activities such as studying, working, or face-to-face social interaction. The recognition of

attention as a scarce resource has given rise to what is now widely referred to as the ‘attention economy.’ Davenport and Beck (2001) define the attention economy as an environment in which organisations compete to attract, retain, and monetise human attention, recognising it as one of the most valuable yet limited resources. In traditional markets, firms compete for consumers’ income and purchasing decisions. However, in digital markets, the primary objective of competition is to increase users’ attention. Building on this idea, Wu (2019) argues that many digital platforms operate on business models that convert user attention into economic value through advertising and data-driven monetisation. Consequently, success in the digital economy is determined not only by the quantity of available information, but also by the ability to capture and sustain users’ limited attention. As Hari (2023) further observes, modern digital platforms are intentionally designed to maximise user engagement through personalised recommendations, continuous content delivery, and persuasive interface designs, making attention the central currency in today’s digital ecosystem. These platforms also employ persuasive techniques like variable rewards to sustain users’ attention for extended periods (Alter, 2017; Bhargava & Velasquez, 2021). Consequently, the amount of attention consumed by digital media is unlikely to depend solely on personal preferences but is also shaped by behavioural and psychological factors.

Within this environment, Gen-Z represents one of the most active participants in the attention economy. Having grown up with smartphones and social media, they are exposed to continuous digital stimulation from an early age, making their attention particularly susceptible to competing digital demands. In this regard, different behavioural and psychological factors influence the extent to which individuals devote their attention to digital media. While existing studies have examined these factors independently, relatively a handful of research has investigated their combined influence by explicitly treating attention as a scarce economic resource. Against this backdrop, our study adopts a microeconomic perspective of attention scarcity to examine how factors like attention fragmentation, fear of missing out (FOMO), and attention awareness influence digital media consumption among Gen-Z in Assam.

2. Methodology

2.1 Research Design and Sample Size

We employ a quantitative approach to assess the factors influencing the attention consumption of Gen-Z in Assam. Assam is selected as the study area because it is the largest state in Northeast India by population and comprises a diverse youth population with increasing access to smartphones, internet connectivity, and digital media platforms. Accordingly, primary data are collected from 150 Gen-Z respondents (aged 18-26 years) residing or studying in different districts of Assam through a structured questionnaire.

2.2 Conceptual Framework

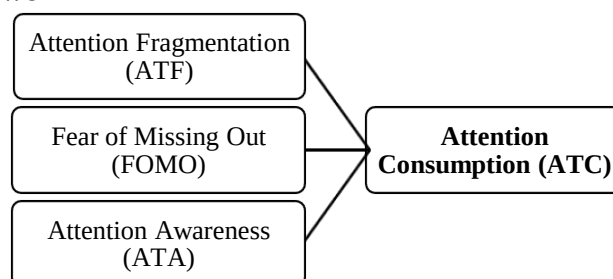


Figure 1: Conceptual Framework

Source: Authors' construction

In our study, as shown in Figure 1, the dependent variable, attention consumption (ATC), refers to the amount and quality of cognitive attention that individuals allocate to digital content across different media platforms. As digital natives, Gen-Z consumes information through multiple platforms simultaneously, often shifting rapidly between applications, formats, and content types. This pattern of media use has transformed not only the quantity of attention devoted to digital content but also the way attention is distributed across competing sources of information (Przybylski & Weinstein, 2017). Research suggests that frequent exposure to personalised and rapidly changing digital content encourages fragmented attention, shorter engagement cycles, and habitual checking behaviours, making attention a central behavioural outcome (Montag & Hegelich, 2020).

We then consider three independent variables. The first variable, attention fragmentation (ATF), refers to the division of an individual's cognitive focus across multiple sources of information, tasks, or digital stimuli, resulting in frequent shifts in attention rather than sustained concentration on a single activity. In digital environments, users are constantly exposed to notifications, short-form videos, algorithmically curated feeds, and multiple simultaneously accessible platforms, all of which encourage rapid switching between content. Rather than maintaining prolonged engagement, individuals allocate their attention in brief, discontinuous episodes, making fragmented attention a defining characteristic of contemporary digital behaviour (Fournier et al., 2026). Chen et al. (2022) argue that individuals addicted to short-form videos show greater distraction and poor attentional concentration. The reason behind this may be attributed to the modern digital environment that promotes fragmented attention and frequent task switching (Mark, 2023). Supporting these findings, Poles (2025) has observed how the intensity of social media use has further led to shorter attention spans, fragmented memory recall, and feelings of overstimulation. This suggests that ATF is associated with greater attention consumption. In addition to ATF, we consider fear of missing out (FOMO) as our second independent variable. FOMO is conceptualised as a social anxiety stemming from the belief that others are having rewarding experiences in one's absence (Przybylski et al., 2013). Fioravanti et al. (2021) and Gupta et al. (2022) have found a significant positive relationship between attention consumption due to social media addiction and FOMO, indicating that individuals who are more addicted to social media are more susceptible to feelings of FOMO. Furthermore, a strong positive association is found between FOMO and university students' attention consumption because of social media engagement (Elsayed, 2025). Finally, we consider Attention Awareness (ATA), which measures the extent to which individuals are aware of attention as a valuable resource and their perception of the strategies employed by digital platforms to capture and retain user engagement. AlYousef et al. (2025) observe that social media addiction is linked to lower mindful attention among university students. This is because high digital media use weakens individuals' attention awareness. Broadening this perspective, Xie et al. (2023) discussed the concept of attention control as a cognitive process responsible for regulating the allocation and focus of attention. They also discussed that increased procrastination is not only directly affected by addiction to short-form videos but also indirectly affected by impaired control of attentional consumption.

2.3 Model Specification

Based on the conceptual framework and the preceding empirical evidence, we examine how ATF, FOMO, and ATA influence ATC of Gen-Z in Assam. To empirically estimate these relationships, a multivariate regression model is specified as follows:

$$ATC_i = \beta_0 + \beta_1(ATF_i) + \beta_2(FOMO_i) + \beta_3(ATA_i) + \varepsilon_i \quad (1)$$

The regression model is estimated using the ordinary least squares (OLS) method. Diagnostic tests are then carried out. To determine whether multicollinearity is present, we examine the variance inflation factor (VIF). If the VIF value is below the threshold of 10, multicollinearity is not a concern. Furthermore, to assess heteroskedasticity, both Breusch-Pagan and White's tests are undertaken. A non-significant χ^2 test statistic (p-value > 0.05) indicates the validity of our models' standard errors.

2.4 Hypotheses

For empirical investigation, the study proposed the following null hypotheses:

H₀₁: ATF has no significant influence on ATC among Gen-Z respondents.

H₀₂: FOMO has no significant influence on ATC among Gen-Z respondents.

H₀₃: ATA has no significant influence on ATC among Gen-Z respondents.

3. Results and Discussion

Table 1: Demographic Characteristics and Daily Attention Consumption of the Respondents

Characteristics	Categories	Frequency
Gender	Male	65
	Female	85
Educational level	Undergraduate Student	125
	Postgraduate Student	15
	Working Professional	5
	Other	5
Daily time spent on social media	< 1 hour	5
	1-2 hours	17
	2-4 hours	67
	4-6 hours	36
	> 6 hours	25

Source: Authors' estimation

Table 1 shows that the study is undertaken by considering 150 respondents, out of which 56 percent (84) are female. The educational status shows that among the respondents, the majority of the respondents (125) are undergraduate Gen-Z students. In addition to that, the largest proportion of respondents (44.7 percent) have reported spending 2-4 hours on social media a day, followed by 24.0 percent spent 4-6 hours. Only 3.3 percent have reported using social media for less than 1 hour daily. This indicates that the majority of the Gen-Z respondents spend at least two hours on average per day on social media, implying a high level of attention consumption by digital media engagement.

Table 2: Results of Multivariate Regression Analysis

Variables	Coefficients
ATF	0.926***
FOMO	0.256
ATA	-0.155

Model Summary	
R ²	0.613
Adj-R ²	0.597
Mean VIF	1.358
White Test	$\chi^2 = 7.710$ (p = 0.260)
Breusch-Pagan Test	$\chi^2 = 4.674$ (p = 0.197)

Note: *** indicates level of significance at 1 percent

Table 2 presents the results of our multivariate regression analysis. The regression estimates show that ATF ($\beta = 0.926$, $p < 0.001$) exerts a positive and statistically significant influence on ATC. This supports the broader argument of the attention economy theory, which views human attention as a scarce resource that digital platforms actively compete to capture and monetise (Davenport & Beck, 2001). The finding is supported by Bayaskar and Sharma (2026), who have found a significant bivariate association between ATF and ATC due to social media use. Similarly, according to Raut (2024), young adults have fragmented and entertainment-oriented attention profiles, with an excessive amount of attention given to social media and entertainment platforms. The result is also consistent with the limited capacity model of motivated mediated message processing (LC4MP) proposed by Lang (2017), which suggests that individuals possess finite cognitive resources for processing information. As digital platforms continuously present novel and competing stimuli, users repeatedly reallocate their limited attention across different pieces of content, increasing overall attention consumption. Similar observations have been reported by Bedeley et al. (2025) and Bayaskar and Sharma (2026), who argue that digital platforms exploit cognitive biases to sustain prolonged user engagement, while rapidly changing online content reshapes attention into shorter, fragmented cycles rather than sustained concentration. In contrast to ATF, FOMO and ATA, even though they exhibit theoretically consistent signs, are not statistically significant. This finding contrasts with previous studies that identified FOMO as a significant predictor of attention consumption (Argan et al., 2022; Tuna et al., 2025). Similarly, the insignificant association of ATA with AFC resonates with the distinction between awareness and self-regulation in behavioural research, where knowledge of a behaviour may not automatically lead to behavioural change (Bandura, 1991). Consequently, we can infer that our null hypothesis H_{01} can be rejected, whereas H_{02} and H_{03} can be accepted.

Diagnostic tests further reveal that both of our considered models have moderately high R² and adjusted-R² values, indicating their strong explanatory power. In addition, both models are free from multicollinearity and heteroscedasticity.

4. Conclusion

Our study reinforces the view that attention is a scarce economic resource which is increasingly contested in information-rich digital media environments. The findings identify ATF as the most significant predictor of ATC, while FOMO and ATA do not exhibit statistically significant effects among Gen-Z. This study offers two important implications. Theoretically, it broadens the literature on the attention economy by showing that attention consumption is better understood through observable behavioural patterns rather than solely through psychological motivations. This shifts the focus from explaining ‘why’ individuals engage with digital platforms to understanding ‘how’ digital environments influence the distribution of cognitive resources. Practically, the findings suggest that initiatives aimed at promoting digital well-being should move beyond awareness-based campaigns and instead emphasise interventions

that encourage sustained attention and reduce fragmented patterns of digital engagement. Such evidence can inform educational institutions in designing digital literacy programmes, assist policymakers in developing healthier digital engagement strategies for young people, and encourage technology developers to consider attention-preserving design principles alongside user engagement metrics. As artificial intelligence (AI) is now increasingly shaping the way digital content is created, recommended, and consumed, these interventions become even more critical. Emerging global initiatives, such as the European Union's Digital Services Act, which promotes greater transparency and accountability in recommender systems, reflect a growing recognition that technological innovation must be accompanied by safeguards for users' cognitive well-being. Complementing such measures with attention-centred digital literacy and human-centred platform design can help ensure that AI enhances, rather than fragments, users' cognitive engagement. Ultimately, sustaining human attention in the age of AI will require a shared commitment from educators, policymakers, researchers, and technology developers to create digital ecosystems that value attention not merely as a commodity, but as a fundamental resource for learning, productivity, and well-being.

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